

Solutions — Practice Midterm (Alt A)

Problem 1

(a) All eight faces appear exactly once in eight rolls:

$$\frac{8!}{8^8} \approx 0.0024032593.$$

(b) At least one 8 appears:

$$P(\geq 1 \text{ eight}) = 1 - \left(\frac{7}{8}\right)^8 \approx 0.6563910842.$$

Problem 2

Let I be the event of identical twins and GG the sonogram event. Then

$$P(I | GG) = \frac{P(I)P(GG | I)}{P(I)P(GG | I) + P(I^c)P(GG | I^c)} = \frac{0.4 \cdot \frac{1}{2}}{0.4 \cdot \frac{1}{2} + 0.6 \cdot \frac{1}{4}} = \frac{4}{7} \approx 0.5714.$$

Problem 3

Let $A \in \{0.28, 0.09, -0.10\}$ and $B \in \{0.15, -0.08, 0.08\}$, each equally likely.

$$E[A] = 0.09, \quad E[B] = 0.05.$$

$$E[A^2] = 0.0321667, \quad E[B^2] = 0.0117667.$$

$$\text{Var}(A) = 0.0240667, \quad \text{Var}(B) = 0.0092667.$$

$$E[AB] = 0.0089333 \Rightarrow \text{Cov}(A, B) = E[AB] - E[A]E[B] = 0.0044333.$$

$$\text{Corr}(A, B) = \frac{\text{Cov}(A, B)}{\sqrt{\text{Var}(A)\text{Var}(B)}} \approx 0.2969.$$

For a 50/50 portfolio,

$$E[P] = 0.5E[A] + 0.5E[B] = 0.07, \quad \text{Var}(P) = 0.25(\text{Var}(A) + \text{Var}(B) + 2\text{Cov}(A, B)) = 0.01055,$$

so $\sigma_P = \sqrt{0.01055} \approx 0.1027$.

Problem 4

With $X \sim \text{Bin}(82, 0.93)$,

$$P(X \leq 80) = 1 - P(X = 81) - P(X = 82) = 1 - \binom{82}{81}(0.93)^{81}(0.07) - \binom{82}{82}(0.93)^{82} \approx 0.9813.$$

Problem 5

For i.i.d. continuous variables, $P(\text{record at } i) = 1/i$. Hence

$$E[Y_n] = \sum_{i=1}^n \frac{1}{i}.$$

For $n = 2$: $\text{Var}(Y_2) = \text{Var}(1 + I_2) = \text{Var}(I_2) = \frac{1}{2}(1 - \frac{1}{2}) = \frac{1}{4}$.

For $n = 3$: Using $E[I_2 I_3] = P(X_1 > X_2 > X_3) = 1/6$,

$$\text{Cov}(I_2, I_3) = \frac{1}{6} - \frac{1}{2} \cdot \frac{1}{3} = 0,$$

so $\text{Var}(Y_3) = \frac{1}{4} + \frac{2}{9} = \frac{17}{36} \approx 0.4722$.

Problem 6

Profit is

$$\Pi(x) = \begin{cases} -c(x), & 7 \leq x < 7.5, \\ 2.40 - c(x), & x \geq 7.5, \end{cases} \quad c(x) = \frac{x}{12} + 0.30.$$

With $f(x)$ triangular on $[7, 9]$ as given,

$$E[\Pi] = -\int_7^{7.5} c(x)(x-7) dx + \int_{7.5}^8 (2.40 - c(x))(x-7) dx + \int_8^9 (2.40 - c(x))(9-x) dx = \frac{17}{15} \approx 1.1333.$$